



SHOP BILLING SYSTEM Using A Thermal Printer

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This shop billing system is a comprehensive solution aimed at streamlining transaction management within a retail environment. Combining Python for graphical user interface (GUI) functionality and Arduino for controlling a thermal printer (RP203), this innovative system enables users to input customer details, select items from an inventory, manage a cart, and seamlessly print bills. Key features include customer information input for personalised billing, inventory management with options for selecting items, specifying quantities, and choosing units of measurement.

The system facilitates cart management by displaying added items, enabling easy item deletion, and instant total amount calculation. It also handles unknown items, allowing for their addition with automatic amount calculation. The system architecture consists of a Python GUI developed using Tkinter for user interaction and Arduino interfacing with the thermal printer for bill printing, creating a robust and efficient transaction management solution for retail settings. To set it

```
import serial
import tkinter as tk
from tkinter import ttk, simpledialog, messagebox

class GrocerySystem:
    def __init__(self, root):
        # Initialize the main window
        self.root = root
        self.root.title("Grocery System")

        # Initialize a dictionary to store cart items
        self.items = {}

        # Initialize customer information
        self.name = ""
        self.phone = ""

        # Initialize the inventory with sample items
        self.inventory = {
            "Dal": {"gram_price": 3.99, "kg_price": 39.99},
            "Rice": {"gram_price": 2.49, "kg_price": 24.99},
            "Wheat": {"gram_price": 1.99, "kg_price": 19.99},
        }

        # Configure ttk style
        self.style = ttk.Style()
        self.style.configure("TButton", font=("Helvetica", 12))
```

Fig. 1: GUI code snippet

BILL OF MATERIALS		
Components	Description	Quantity
Arduino Uno (MOD1)	Development board	1
USB Type C	Adaptor	1
RP203 (MOD 2)	Thermal printer	1
12V, 1A power adaptor	Power source for thermal printer	1
Jumper wires	Male to female	6

Note: Most thermal printers come with a printer page, but if it does not, you need to add that to the shopping cart as well.

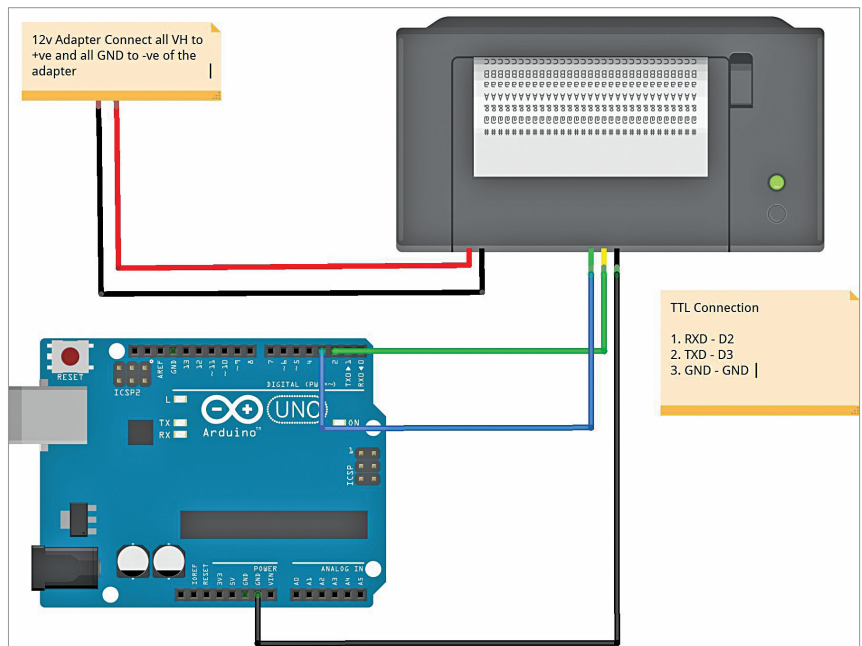


Fig. 2: Circuit diagram